

Physics as the Topology of a Non-Invertible Projection

Axiomatic Core of the Quantum Gauge Theory of Time

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Abstract

$$\partial_I J^I = 0, \quad I = 1, \dots, 5, \quad \text{under restricted accessibility.}$$

This single condition is the minimal physical input required to complete the pre-physical closure established by the finite-rank non-invertible framework [1, 2, 3, 4]: a domain with no intrinsic ordering, a truncated observability metric, and a non-trivial inaccessible sector. Imposed on that structure, it forces by necessity a compact zero mode $\chi \in S^1$, the self-conjugate Majorana bulk field $\Xi = C\bar{\Xi}$, the non-invertible rank-5 projection Π , and the minimal representational structure $\mathbb{R}^4 \times S^1(\tau)$, not postulated as a five-dimensional manifold but forced as the unique domain supporting non-degenerate boundary observables, with golden temporal connection $\Gamma_\tau = \varphi/\sqrt{5}$.

On $\mathbb{R}^4 \times S^1(\tau)$, the conservation law decomposes as $\partial_\tau J^\tau + \partial_\mu J^\mu = 0$, where $J^\tau = \rho^2 \Gamma_\tau$ is the coherent temporal flux and $\partial_\mu J^\mu$ the entropic spatial divergence. Restricted accessibility forces $\ker(\Pi) \neq 0$: the zero mode cannot cross completely. What does not cross returns as boundary tension $V'(|\Psi|)$, where $\Psi = \Pi\Xi$, and as the discrete residue that organises the observable world.

From this residue — through the companion derivations of the series [7, 8, 10, 11, 15, 16] — follow as structural invariants, not postulates: $\alpha^{-1} = (5 + 3\sqrt{5})^2$, $G = U(1) \times SU(2) \times SU(3)$, $\sin^2 \theta_W = 3/13$, $6Q \in \mathbb{Z}$, the breathing law $\mathcal{B}_{k+1} = 2\mathcal{B}_k^{2/3}$, and time as the irreversible ordering of incomplete crossing. One external dimensional anchor is required, R_∞ [5], from which the dimensional cascade closes [15].

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<https://pasqualecamelia.github.io>

Introduction: The Zero Mode

$$\partial_I J^I = 0 \text{ (restricted accessibility)} \implies \ker(\Pi) \neq 0 \implies \text{physics.}$$

There is a zero mode.

Not a particle. Not a field excitation. Not a coordinate. A compact internal phase $\chi \in S^1$ — the minimal topological structure that a globally conserved flow must carry in order to recognise itself after transport. It closes. It is coherent. It carries the complete phase information of the Noetherian symmetry.

The zero mode is the anterior structure of the bulk. Not the representational manifold $\mathcal{M}^5 = \mathbb{R}^4 \times S^1(\tau)$ — that comes later, as a derived representational necessity. The zero mode is the anterior thing: the structural coherence that the conservation law requires in order to remain globally consistent under restricted accessibility.

At the projection interface, the zero mode becomes evanescent, loses dimensionality, collapses onto its surface. Full coherence on one side. Entropic decay at the interface. The internal generative phase, the circle that closes on itself, does not cross.

What does not cross does not disappear. It returns as tension in the boundary structure. Not an interaction added from outside. The exact measure of what the zero mode could not carry through.

The conservation law on $\mathcal{M}^5$ decomposes as:

$$\partial_\tau J^\tau + \partial_\mu J^\mu = 0,$$

where $J^\tau = \rho^2 \Gamma_\tau$ is the coherent temporal flux and $\partial_\mu J^\mu$ is the entropic spatial divergence. This is not the sum of two terms that cancel independently. It is a balance: the coherence that the temporal fibre loses, the boundary gains — and vice versa. The departure from that balance, σ , is the measure of physics: $\sigma = 0$ is pure coherence; $\sigma > 0$ is matter and complexity; $\sigma \rightarrow \infty$ is the entropic floor. All of physics is the decomposition of one conservation law.

The bulk Majorana field $\Xi = C\bar{\Xi}$ is self-conjugate, without charge, without internal distinction. Its equation in the bulk is linear:

$$\mathcal{D}_{\text{bulk}} \Xi = 0.$$

Linearity is not simplicity. It is the absence of distinction. The bulk may contain infinitely many modes, but as long as they remain physically undifferentiated — no selection, no charge asymmetry, no observable hierarchy — their collective effect is a closed linear dynamics. The Majorana condition makes that absence exact.

The boundary is where difference appears. The projection has broken the indifference of the bulk: it has selected modes, created asymmetries, introduced a kernel. And since $\ker(\Pi) \neq \{0\}$, the projected field $\Psi = \Pi\Xi$ cannot satisfy a closed linear equation. At the boundary on $\mathbb{R}^4$, the projected field obeys:

$$(\gamma^\mu D_\mu + \alpha\xi + i\alpha\eta\gamma^5)\Psi = V'(|\Psi|)\Psi.$$

The left-hand side governs the two chiral channels into which the indifference of the bulk has broken: the longitudinal channel ξ gives gravity, the transverse channel η gives electromagnetism. Two forces. One equation. Two chiral readings of the same fact. The right-hand side is not added from outside — it is the return tension of the phase that did not cross, the exact echo of the evanescent zero mode.

A zero mode that fails to cross the interface completely leaves, as its residue, everything we call physics.

1 The Axiom and Its Decomposition

The pre-physical analyses [1, 2, 3, 4] establish that observational access to a globally coherent structure is necessarily mediated by a non-invertible finite-rank projection, and that no intrinsic ordering is available within the resulting closed representational domain. In this paper we introduce the minimal physical input that completes this structure: a globally conserved Noetherian flow.

Axiom 1 (Noetherian conservation under restricted accessibility). There exists a globally conserved Noetherian flow

$$\partial_I J^I = 0, \quad I = 1, \dots, 5, \quad (1)$$

under conditions of restricted accessibility: not all invariants of J^I are operationally available to a reduced boundary observer. This is the minimal physical input that uniquely completes the pre-physical structure [8].

Here $J^I = (J^\tau, J^\mu)$ is the five-component Noether current of the conserved flow on $\mathcal{M}^5$: $J^\tau = \rho^2 \Gamma_\tau$ is the coherent temporal flux, where $\rho = |\Xi|$ is the bulk field amplitude and $\Gamma_\tau = \varphi/\sqrt{5}$ is the golden temporal connection; J^μ ($\mu = 1, 2, 3, 4$) is the entropic spatial divergence. The index $I = 1, \dots, 5$ runs over the full bulk directions of $\mathcal{M}^5 = \mathbb{R}^4 \times S^1(\tau)$. The condition $\partial_I J^I = 0$ is the global conservation law that closes the bulk coherence.

As Theorem 4 below establishes, the minimal domain supporting non-degenerate reduced bilinear invariants is $\mathcal{M}^5 = \mathbb{R}^4 \times S^1(\tau)$. On this manifold the conservation law decomposes as

$$\partial_\tau J^\tau + \partial_\mu J^\mu = 0 \implies \sigma \equiv -(\partial_\tau J^\tau + \partial_\mu J^\mu)|_{\partial V}, \quad (2)$$

where $J^\tau = \rho^2 \Gamma_\tau$ is the coherent temporal flux and σ measures the observable departure at the boundary [7]. This is not the vanishing of two independent terms. It is a balance: whatever coherence the temporal fibre loses, the spatial boundary gains — and vice versa. In the full bulk $\sigma = 0$ exactly, by the axiom. At the boundary, restricted accessibility makes σ observable: it measures the coarse-grained departure from balance that the projection Π cannot hide. When $\sigma = 0$: pure coherence, gauge active. When $\sigma > 0$: constructive imbalance, forcing complexity at levels $n = 3, 2$ ([13], §9.3). When $\sigma \gg 0$: dissipation, Boltzmann statistics. When $\sigma \rightarrow \infty$: the entropic floor, the CMB. *All of physics is the decomposition of one conservation law.*

The non-invertible projection $\Pi : \mathcal{M}^5 \rightarrow \mathbb{R}^4$ and the projected field $\Psi = \Pi\Xi$ are defined precisely in (4) below, together with the Pincherle filter that realises Π and fixes the fine-structure constant α as its unique pole.

Theorem 1 (The zero mode is necessary). *Under Axiom 1, global conservation requires a compact internal phase $\chi \in S^1$ — a primordial zero mode — whose carrier must be double-valued under 2π transport:*

$$\Xi(\chi + 2\pi) = -\Xi(\chi), \quad \Xi = C\bar{\Xi}. \quad (3)$$

A single-valued carrier, $\Xi(\chi + 2\pi) = +\Xi(\chi)$, has trivial holonomy: its only projection invariant is the identity, which carries no observable information [6]. Double-valuedness is therefore not a postulate — it is the minimal topological condition for the zero mode to produce a non-trivial boundary algebra under Π ([6], Proposition 5.1).

Proof. A single-valued carrier $\Xi(\chi + 2\pi) = +\Xi(\chi)$ has holonomy operator $\hat{H} = +1$: its only projection invariant is the identity, generating no observable structure under Π . Double-valuedness gives $\hat{H} = -1$, with minimal polynomial $z+1 = 0$ over $\mathbb{Z}$: non-trivial, producing a two-dimensional boundary algebra. The self-conjugacy $\Xi = C\bar{\Xi}$ is the unique constraint compatible with global charge neutrality under a single conserved flow: any other choice requires either a pre-existing metric or an independent charge sector, neither available at this structural level. $\square$ $\square$

Remark 2 (Majorana as the unique admissible carrier). The identification $\Xi = C\bar{\Xi}$ follows from elimination of all alternatives, proved by exhaustion in [6], §§5–6.

- **Vector fields** (A_μ) require a pre-existing metric to define index contraction, but the metric must itself emerge from the projection; assuming it as input is circular.
- **Dirac spinors** carry a complex charge ($\psi \neq C\bar{\psi}$): the total bulk charge does not vanish, violating global neutrality under a single conserved flow.
- **Weyl spinors** are chiral and globally incomplete on a compact fibre $S^1(\tau)$: the two Weyl sectors are independent, and the self-referential closure required by Axiom 1 is absent.

A Majorana spinor is real, globally neutral, self-conjugate under C , and generates all observable structures as bilinears. It is the unique carrier satisfying all four requirements simultaneously.

Theorem 3 (The interface is necessary). *Under Axiom 1, restricted accessibility forces a non-invertible projection $\Pi : \mathcal{M}^5 \rightarrow \mathbb{R}^4$ with $\ker(\Pi) \neq \{0\}$. The rank is uniquely $r = 5$: it is the only value for which the holonomy norm equation $|O|^2 = F_r = r$ has a solution, as proved in §3 below ([4], Theorem 3.1; [11], Theorem 2). At this interface the zero mode becomes evanescent: its phase information does not propagate into the observable sector. What does not propagate returns as boundary non-linearity $V'(|\Psi|)$, where $\Psi = \Pi\Xi$ ([6], Section 10).*

Proof. **Why $\ker(\Pi) \neq \{0\}$.** Restricted accessibility means the boundary observer cannot reconstruct all bulk invariants. A bijective Π would allow full reconstruction; hence Π must be non-invertible, forcing $\ker(\Pi) \neq \{0\}$ [3].

Why $r = 5$ uniquely. The Majorana double-valuedness forces the holonomy norm $|O_n|^2 = F_n$ (the n -th Fibonacci number), as the Majorana condition eliminates all cross-terms in the norm (proved in the Theorem 9 below). For the rank r to equal its own

holonomy norm, we need $F_r = r$. Among all positive integers: $F_1 = 1 = 1 \checkmark$, $F_2 = 1 \neq 2$, $F_3 = 2 \neq 3$, $F_4 = 3 \neq 4$, $F_5 = 5 = 5 \checkmark$, $F_r < r$ for all $r > 5$. The non-trivial solution is unique: $r = 5$. Primality of $r = 5$ rigidifies the selection: it is the unique prime p with $F_p = p$ non-trivially ([4], Theorem 3.1). $\square$ $\square$

Theorem 4 (The manifold and the golden scale are necessary). *Under Axiom 1, the minimal domain supporting non-degenerate reduced bilinear invariants is $\mathcal{M}^5 = \mathbb{R}^4 \times S^1(\tau)$. The second-order non-Abelian closure forces spectral separation $\sqrt{5}$, fixing the golden temporal connection:*

$$\Gamma_\tau = \frac{\varphi}{\sqrt{5}}, \quad \varphi = \frac{1 + \sqrt{5}}{2}.$$

The boundary carries a discrete conformal structure parametrised by φ , not a metric ([6], Sections 11–12; [16], §2).

Remark 5 (Uniqueness of $\mathcal{M}^5 = \mathbb{R}^4 \times S^1(\tau)$). The factorised product structure is forced, not assumed. The argument is by exhaustion, developed in [6], §§10–11 and [13], Step 3.

- **Four extended dimensions.** The projection Π must act on a domain that supports all projected bilinear observables without pre-assuming a metric. This requires at least four extended directions; a compact extended domain would enforce global identifications among projected observables, contradicting restricted accessibility. Hence $\mathbb{R}^4$ is the unique non-compact four-dimensional factor.
- **One compact fibre.** The zero mode $\chi \in S^1$ requires exactly one compact internal direction along which global phase transport closes. Additional compact directions would introduce independent conserved channels, violating the single-flow condition of Axiom 1. Hence $S^1(\tau)$ is the unique compact factor.
- **Trivial fibration.** Twisted bundles and non-trivial fibrations correlate internal phase accessibility with projected observables, violating the enforced separation between $\ker(\Pi)$ and the boundary readout.

The total manifold therefore factorises as $\mathcal{M}^5 = \mathbb{R}^4 \times S^1(\tau)$, the minimal structure that exists at all under the given constraints.

No framework intrinsically formulated on $\mathbb{R}^4$ can derive the algebra of admissible observables, dimensionless constants as structural invariants, or the bulk-to-boundary split — as proved in [3], Theorems 1–4, and in [5], Theorem 2.1, without any assumption on field content. The projection Π with non-trivial kernel is therefore not an additional hypothesis: it is the only structure consistent with these no-go results. The kernel is the boundary echo of the evanescent zero mode — the information that did not cross becomes the information that cannot be self-derived.

Definition 6 (Projection operator and projected field). The *non-invertible projection* $\Pi : \mathcal{M}^5 \rightarrow \mathbb{R}^4$ maps the bulk Majorana spinor to the observable boundary field. It satisfies $\ker(\Pi) \neq \{0\}$: information is irreversibly lost. Its truncated singular value decomposition retains exactly five modes:

$$\Psi := \Pi \Xi, \quad \Pi = U \Sigma_5 V^\dagger, \quad \Sigma_5 = \text{diag}(\sigma_1, \dots, \sigma_5, 0, \dots), \quad (4)$$

where $U = [u_1 | \dots | u_5]$ and $V^\dagger = [v_1 | \dots | v_5]^\dagger$ are the orthonormal SVD bases of $\mathcal{H}_{\text{obs}}$ and $\mathcal{H}_{\text{bulk}}$ respectively, whose columns span the Mellin-wavelet subspace $\{1, \log x, (\log x)^2, (\log x)^3, (\log x)^4\}$ ([1], §9; [2], Appendix A.3), and $\sigma_k \propto \varphi^{-k}$ are the golden singular values [2, 4]. The Pincherle dilatation $D_\tau = \tau \partial_\tau$ on $S^1(\tau)$, with fixed point $\lambda = \varphi$, realises Π as the rank-5 rational filter $\mathcal{H}(z) = P_4(z)/(1 - \alpha z)$, $z = \varphi^{-1}$ [1], whose pole α^{-1} is the *fine-structure constant* $\alpha \approx 1/137$: the unique pole compatible with the autosimilar bulk [2]. $\Psi \in L^2(\mathbb{R}^4)$ is the only field operationally accessible to a boundary observer.

$$\begin{aligned} \partial_I J^I = 0 &\implies \chi \in S^1 \text{ (zero mode)} \implies \\ \Xi = C\bar{\Xi}, \Pi \text{ rank-5, } \mathcal{M}^5, \Gamma_\tau &= \varphi/\sqrt{5}. \end{aligned}$$

2 The Interface and Its Tension

In the bulk $\mathcal{M}^5$, the Majorana spinor Ξ satisfies a linear bulk Dirac equation $\mathcal{D}_{\text{bulk}} \Xi = 0$. After projection to $\mathbb{R}^4$, the field $\Psi = \Pi \Xi$ inherits the Dirac structure on $\mathbb{R}^4$ — but $\ker(\Pi) \neq \{0\}$ (Theorem 3) makes a closed linear equation impossible. The projection residue returns as a non-linear term. The boundary field equation on $\mathbb{R}^4$ is:

$$(\gamma^\mu D_\mu + \alpha \xi + i\alpha \eta \gamma^5) \Psi = V'(|\Psi|) \Psi, \quad (5)$$

where $V'(|\Psi|) \Psi$ encodes the return of the phase that did not cross the projection. The non-linearity is not added from outside. It is the exact measure of what the zero mode could not carry through [9, 13].

The two limits separate the two forces:

$$\eta = 0 : (\gamma^\mu D_\mu + \alpha \xi) \Psi = 0 \text{ [gravity]}, \quad \xi = 0 : (\gamma^\mu D_\mu + i\alpha \eta \gamma^5) \Psi = 0 \text{ [EM]}. \quad (6)$$

Two forces. One equation. Two chiral readings of the same fact. Electromagnetism and gravitation are not independent fields to be unified. They are the two chiral boundary readouts of a single Majorana bulk [13]. Lorentz covariance is inherited from the spinorial structure of Π , not postulated.

The boundary metric $K = \Pi^\dagger \Pi$ arises from truncated SVD. The coupling invariant is structurally fixed:

$$\mathcal{K}_\tau = \xi \eta = \alpha^{-1}. \quad (7)$$

This is not a free parameter. It is the product of the two curvature components of the temporal fibration, determined by the truncated SVD without external input ([2], [4]).

The two components evolve by hyperbolic rotation on the invariant surface $\mathcal{K}_\tau$:

$$\dot{\xi} = -\Gamma_\tau \xi, \quad \dot{\eta} = +\Gamma_\tau \eta, \quad \Gamma_\tau = \frac{\varphi}{\sqrt{5}}. \quad (8)$$

The field breathes: ξ decays and η grows (or vice versa) while their product is conserved. What changes is the logarithmic ratio $\theta_{\log} = \frac{1}{2} \log(\eta/\xi)$ — and that logarithmic phase is time [7, 15].

The boundary undergoes coherence-radiation exchange. The boundary amplitude at cycle k is:

$$\mathcal{B}_k := |\Psi_k|^2, \quad \mathcal{B}_{k+1} = 2\mathcal{B}_k^{2/3}, \quad \mathcal{B}^* = 8. \quad (9)$$

Each cycle is the boundary's attempt to absorb the zero mode tension; each cycle leaves a residue; the accumulated residue forces $6Q \in \mathbb{Z}$ ([7, 15]). The framework is the minimal extension compatible with finite-rank observability under a single conserved zero mode; it is unique up to rank ([8]).

$$\ker(\Pi) \neq 0 \implies V'(|\Psi|) \text{ (boundary tension)} \implies \mathcal{B}_{k+1} = 2\mathcal{B}_k^{2/3}, \quad 6Q \in \mathbb{Z}.$$

3 The Residue and Its Algebra

The zero mode that does not cross leaves a discrete invariant structure. That structure is not arithmetic: it is the topology of a Fibonacci holonomy trajectory. The Majorana spinor transported by $\Gamma_\tau = \varphi/\sqrt{5}$ along $S^1(\tau)$ traces a trajectory whose stable nodes are the Fibonacci numbers ($\Gamma_\tau^k \approx F_k/\sqrt{5}$, Binet formula). The admissible modes

$$\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$$

are the nodes of minimal holonomy of an irrational rotation on a rank-5 surface [1]. The numbers $\{2, 3, 5, 7, 11\}$, 137, 54, $3/13$ are not arithmetic choices: they are the holonomy invariants of the spinor's Fibonacci trajectory, expressed in the ring $\mathbb{Z}[\sqrt{5}]$.

The four results below are one result in four coordinate systems ([11], Theorem 5.1; [13], Theorem 8.2): $\alpha^{-1} = (5 + 3\sqrt{5})^2 \iff \sin^2 \theta_W = 3/13$, where $F_5 = |\mathcal{N}_{\text{adm}}| = 5$ is the filter rank and $F_4 = 3$ is the electroweak rank — Fibonacci numbers of the holonomy trajectory on $S^1(\tau)$, not free parameters. $F_4 = \dim \text{Hom}(\tau^{\otimes 6}, \mathbf{1})$ is the dimension of the non-Abelian EW sector; $F_7 = \dim \text{Hom}(\tau^{\otimes 12}, \mathbf{1}) = 13$ is the total fusion multiplicity (the Majorana condition halves $26 \rightarrow 13$) ([11], §8). The full link: $\|v\|^2 = 5\beta + \sigma$, $\delta = \frac{3}{2}\beta$, with $\beta = 36$, $\sigma = 137$, $\delta = 54$, $\|v\|^2 = 317 = 180 + 137$.

Theorem 7 (α^{-1} as the Fibonacci norm of the residue). *The fine-structure constant α — the coupling of electromagnetism, fixed as the pole of the rank-5 Pincherle filter (4) —*

equals the squared norm, in $\mathbb{Z}[\sqrt{5}]$, of the irreducible residue of the evanescent zero mode:

$$\alpha^{-1} = (F_5 + F_4\sqrt{5})^2 = (5 + 3\sqrt{5})^2 = 70 + 30\sqrt{5}, \quad (10)$$

where $F_4 = 3$ and $F_5 = 5$ are consecutive Fibonacci numbers, equal respectively to the electroweak rank and the filter rank of Π .

Proof. $4\varphi^2(\varphi - \frac{1}{2}) = (3 + \sqrt{5})\sqrt{5} = 5 + 3\sqrt{5}$. Square: $(5 + 3\sqrt{5})^2 = 70 + 30\sqrt{5}$. $\square \quad \square$

Theorem 8 (G from discrete closure). $\mathcal{N}_{\text{adm}}$ and the kernel structure of Π determine $G = U(1) \times SU(2) \times SU(3)$ as the unique group compatible with rank-5 boundary metric, breathing attractor $\mathcal{B}^* = 8$, and irreducible residue $\sigma = 137$.

Proof. The $U(1)$ factor is the phase symmetry of the boundary zero mode. $SU(2)$ arises from the two-channel spinorial structure of Π at the lock-in threshold $n = 5$, $k = 2$. $SU(3)$ arises from the confined triplet of the $n = 7$ kernel sector, whose three components cannot be individually projected ([10], Theorem 4.1). Uniqueness follows from the categorical confinement $[5]_q = 0$: no extension of $\mathcal{N}_{\text{adm}}$ beyond five elements is compatible with the boundary algebra ([11], Theorem 2). $\square$

Theorem 9 ($q = e^{i\pi/5}$ is forced). The boundary deformation parameter is uniquely $q = e^{i\pi/5}$, forced by the primary constraints without any free identification.

Proof. • **Spectral norm.** The closure $v_{k+2} = v_{k+1} + v_k$ has minimal polynomial $\lambda^2 - \lambda - 1 = 0$, roots $\lambda_{\pm} = (1 \pm \sqrt{5})/2$, norm $|O|^2 = (\lambda_+ - \lambda_-)^2 = 5$.

- **Holonomy orthogonality.** The holonomy operator $\hat{H}\Xi_n = e^{i\pi/n}\Xi_n$ is self-adjoint; distinct sectors are orthogonal: $\langle \Xi_n, \Xi_m \rangle = 0$ for $n \neq m$.
- **Block-diagonal CM .** M acts on orthogonal factor spaces of $\mathcal{M}^5$, so $[M, \hat{H}] = 0$. The Majorana condition requires $[C, \hat{H}] = 0$: CM is block-diagonal.
- **Fibonacci recurrence.** The Majorana condition forces $|O_1|^2 = |O_2|^2 = 1$. From $\Xi_{k+2} = \Xi_{k+1} + \Xi_k$, cross-terms vanish (block-diagonality + orthogonality), therefore $|O_n|^2 = F_n$.
- **Sector selection.** $F_n = 5$ has unique solution $n = 5$. Primality rigidifies: $r = 5$ is the unique prime with $F_p = p$ non-trivially, forcing $n = r = 5$ and $H^5 = -1$.
- **Intertwining (Perron–Frobenius closure).** Let $A := \Pi \circ T|_{V_{\text{stab}}}$, where $V_{\text{stab}} = \{v : \|\Pi(T^k v)\| \not\rightarrow 0\}$. By Perron–Frobenius (Krein–Rutman), A has spectral radius $r(A) = \varphi$. The intertwining condition $\Pi \circ T = L_q \circ \Pi$ forces $\delta = \varphi$.

- **Branch selection.** $H^5 = -1$ gives candidates $\theta \in \{\pi/5, 3\pi/5, \pi\}$. Only $\theta = \pi/5$ satisfies $2\cos\theta = \varphi$. Therefore $q = e^{i\pi/5}$.

□

□

Proposition 10 (Majorana condition + minimality + normalisation $\Rightarrow a = b = 1 \Rightarrow F_n$). *Let $\Xi_{k+2} = a\Xi_{k+1} + b\Xi_k$ be the most general two-term linear recurrence for the Majorana spinor sectors, with $a, b \in \mathbb{R}_{>0}$. Then:*

- (i) The **Majorana condition** $\Xi = C\bar{\Xi}$ forces the recurrence coefficients to satisfy $a^2 + b^2 \in \mathbb{Z}$ (holonomy integrality on $S^1(\tau)$).
- (ii) The **minimality condition** (unique bulk-to-boundary lift: no redundant modes) forces $ab \leq 1$.
- (iii) The **normalisation condition** $|O_1|^2 = |O_2|^2 = 1$ (boundary zero modes carry unit amplitude) forces $b = 1$ and then $a = 1$.

*Together: $a = b = 1$, giving the Fibonacci recurrence $\Xi_{k+2} = \Xi_{k+1} + \Xi_k$ and $|O_n|^2 = F_n$. **Level A.***

Proof. (i) On the temporal fibre $S^1(\tau)$, each sector Ξ_n acquires holonomy $e^{i\pi/n}$. The Majorana reality condition $C\bar{\Xi}_n = \Xi_n$ forces the holonomy to be unitary with integer-valued norm squared; for a two-term recurrence with positive coefficients, $a^2 + b^2 \in \mathbb{Z}$ is the minimal constraint.

(ii) Minimality (Axiom 1 applied to the lift Π^{-1}) requires that the recurrence does not introduce pairs of modes that project to the same boundary state. For positive a, b , this is equivalent to $ab \leq 1$ (the recurrence has at most one attractor in the forward orbit of Π).

(iii) The boundary zero modes (Jackiw–Rebbi states) normalise to unit amplitude by the boundary metric condition $\langle \Xi_1, \Xi_1 \rangle = \langle \Xi_2, \Xi_2 \rangle = 1$ (proved in P04 [4]). With $|O_1|^2 = 1$ and $|O_2|^2 = 1$, the block-diagonal Majorana structure (Theorem 9) and the integrality constraint $a^2 + b^2 \in \mathbb{Z}$ give $b = 1$. Substituting into the minimality bound $ab \leq 1$ with $b = 1$ gives $a \leq 1$. Integrality then forces $a = 1$. □ □

Proposition 11 (Uniqueness of the minimal construction). *The Fibonacci recurrence $\Xi_{k+2} = \Xi_{k+1} + \Xi_k$ with $|O_1|^2 = |O_2|^2 = 1$ is the **unique** two-term recurrence satisfying all three conditions of Proposition 10 simultaneously. In particular, no other pair (a, b) with $a, b > 0$ produces a boundary-compatible Majorana mode tower on $\mathcal{M}^5$. **Level A.***

Proof. By exhaustion: $a, b \in \mathbb{Z}_{>0}$ (from integrality + minimality). The constraint $ab \leq 1$ with $a, b \geq 1$ forces $a = b = 1$. For non-integer a or b , the holonomy integrality condition fails on the temporal fibre. Uniqueness follows. □ □

Theorem 12 (Electroweak mixing angle). *The boundary algebra $\partial V = \text{TL}_4(\varphi)$ with $q = e^{i\pi/5}$ forces the electroweak mixing angle:*

$$\sin^2 \theta_W = \frac{F_4}{F_7} = \frac{3}{13} \approx 0.2308$$

in agreement with measurement to 0.20%. The explicit isomorphism $\Phi(v_0) = (5 + 3\sqrt{5})^2 = \alpha_{\text{QGT}}^{-1}$ is proved with complete functorial verification [11].

Remark 13 (Derivation of 3/13). The number 3/13 is not fitted. It is the unique ratio forced by the boundary algebra, proved as an unconditional theorem in [11]. The chain is: (i) $\partial V = \text{TL}_4(\varphi)$ with $q = e^{i\pi/5}$ (Theorem 9 above) forces G1–G3 as theorems of the Fibonacci fusion category ([11], Theorem thm:TL4_G123). (ii) G1–G3 imply that the residue $\sigma = \alpha^{-1}$ spans a one-dimensional invariant subspace H_{proj} on which every admissible electroweak transformation acts trivially. The residue is the cost-free mode of the filter $H(z) = P_4(z)/(1 - \alpha z)$ — the pole that passes through the projection unchanged — and therefore does not participate in electroweak symmetry breaking. (iii) The mixing angle is the ratio of the non-Abelian closure cost $\delta = \frac{3}{2}\beta = 54$ to the dynamic denominator $\rho + \delta = 180 + 54 = 234$, where σ is excluded:

$$\sin^2 \theta_W = \frac{\delta}{\rho + \delta} = \frac{54}{234} = \frac{3}{13}.$$

The denominator is 234, not $234 + \sigma = 371$, precisely because $H_{\text{proj}} \cap H_{\text{dyn}} = \{0\}$: the residue and the dynamic sector are orthogonal ([11], Corollary cor:thetaW_global).

Theorem 14 (Categorical confinement). *The quantum integers of $\text{TL}_4(\varphi)$ are palindromic: $[1]_q = 1$, $[2]_q = \varphi$, $[3]_q = \varphi$, $[4]_q = 1$, $[5]_q = 0$. The vanishing $[5]_q = 0$ truncates the algebra at level 5, forcing $|\mathcal{N}_{\text{adm}}| = 5$. The fifth level is not forbidden by hand. It is categorically confined by the algebra that the single axiom forces ([11], Theorem 2).*

Theorem 15 (Time). *Each cycle of $\mathcal{B}_{k+1} = 2\mathcal{B}_k^{2/3}$ carries away a residue of coherence that cannot be recovered. The lost information accumulates monotonically in $\ker \Pi$. Time is not a coordinate. It is the directed loss of zero mode coherence, forced by $\ker(\Pi) \neq 0$ ([7, 8]). Its physical frequency emerges from the global phase-locked loop of the boundary breathing cycle ([15]).*

Proof. At each step k , the boundary state carries amplitude $\mathcal{B}_k = |\Psi_k|^2$. After one cycle of the projection Π and its non-linear return tension $V'(|\Psi|)$, the amplitude obeys $\mathcal{B}_{k+1} = 2\mathcal{B}_k^{2/3}$ (proved in [7], Theorem 1). The factor $2\mathcal{B}_k^{2/3} < \mathcal{B}_k$ for $\mathcal{B}_k > 8$, and $> \mathcal{B}_k$ for $\mathcal{B}_k < 8$: the map contracts toward $\mathcal{B}^* = 8$ from both sides. The Lyapunov exponent at the fixed point is $\lambda = 2/3 < 1$: convergence is monotone and irreversible. At each step, the information in $\ker(\Pi)$ — the part of Ξ that did not project — increases by a fixed

positive amount. This monotone accumulation of inaccessible information is the directed structure that the theory identifies as time. It cannot be reversed: Π is non-invertible, so the accumulated kernel content cannot be projected back. $\square$ $\square$

4 The Consequences

Once the algebra is fixed, the physics follows without additional input.

Theorem 16 (Matter). *The admissible modes $n \in \mathcal{N}_{\text{adm}}$ generate the stable matter sectors. Particles are the winding modes of the zero mode's residue on the rank-5 fibre $S^1(\tau)$ ([12]).*

Theorem 17 (Electromagnetism and gravitation). *Expanding around $\mathcal{B}^* = 8$, the boundary field governs two orthogonal chiral channels of $\Psi = \Pi\Xi$: $\Psi_{\perp} \rightarrow$ Maxwell's equations, $\Psi_{\parallel} \rightarrow$ Einstein gravity. Two forces, one field, two chiral readings of the same zero mode echo. The constants c and $\hbar$ are invariants of the projection geometry, not external inputs ([13]).*

Corollary 18 (Dark matter). *The kernel sector $\ker(\Pi)$ never crosses. It carries no observable charge and no winding modes. Dark matter is not a new species: it is the invisible complement of the observable projection ([14]).*

Theorem 19 ($1/f$ noise as the incoherent limit). *The two structural limits of the interface are: coherent ($|\mathcal{N}_{\text{adm}}| = 5$, α^{-1} and gauge emerge) and incoherent (all modes eliminated, scale-free fluctuations remain: $1/f$ noise). α^{-1} and $1/f$ noise are the two poles of the same interface ([18]).*

The Single Statement

One chain.

$$\underbrace{\partial_I J^I = 0 \implies \chi \in S^1 \implies \ker(\Pi) \neq 0}_{\text{bulk}} \implies \underbrace{V'(|\Psi|) \implies G, \alpha^{-1}, \sin^2 \theta_W, \text{ time.}}_{\text{boundary}}$$

$\mathbb{R}^4$ is not a background assumed in advance. It is the observable boundary structure forced by the projection — the surface that emerges when the internal phase cannot cross completely.

The gauge group $G = U(1) \times SU(2) \times SU(3)$ is forced, not postulated. It is the unique algebra compatible with the rank-5 projection, the boundary metric, and the irreducible residue $\sigma = \alpha^{-1}$.

Time is not an input of the framework. It is the directed residue of incomplete crossing: each cycle carries away something irrecoverable, and that accumulated, irreversible loss is what we call time.

References by Corpus Layer

Suggested reading order: Layer 0 $\rightarrow$ Layer 1 $\rightarrow$ Layer 2 $\rightarrow$ Layer 3. Companion notes provide status, interpretation, and auxiliary closure paths.

References

Layer 0 — Pre-physical closure

- [1] P. Camelia, *Projection-Induced Singular Value Decomposition of Irrational Scale Invariants and the Origin of the Fine-Structure Constant*, QGT Series Paper P01 (2026).
- [2] P. Camelia, *Boundary Metric from Projection-Induced Finite Rank*, QGT Series Paper P02 (2026).
- [3] P. Camelia, *Non-Derivability of Admissible Observables*, QGT Series Paper P03, Zenodo (2026).
- [4] P. Camelia, *Finite-Rank Boundary Metrics*, QGT Series Paper P04, Zenodo (2026).
- [5] P. Camelia, *Structural Constraints on Observable Algebras from Finite-Rank Non-Invertible Projections* (2026).

Layer 1 — Minimal physical closure

- [6] P. Camelia, *A Quantum Gauge Theory of Time (QGT): Foundations from a Noetherian Balance* (2026).
- [7] P. Camelia, *Breathing Equation and Charge Quantisation*, QGT Series Paper P05, Zenodo (2026).
- [8] P. Camelia, *Quantum Mechanics as the Boundary Algebra of a Non-Invertible Projection: Hilbert Space, Born Rule, Measurement, and the Resolution of the Problem of Time*, QGT Series Paper P06, Zenodo (2026).
- [9] P. Camelia, *Field Theory and the Unified Lagrangian*, QGT Series Paper P08, Zenodo (2026).

Layer 2 — Gauge and matter structure

- [10] P. Camelia, *Triplet Structure and Minimal Emergence of $SU(3)$* , QGT Series Paper P07, Zenodo (2026).
- [11] P. Camelia, *Finite-Rank Selection and the Gauge Structure*, QGT Series Paper P09 (2026).
- [12] P. Camelia, *Matter as Admissible Modes*, QGT Series Paper P10, Zenodo (2026).
- [13] P. Camelia, *Electromagnetism and Gravitation as Chiral Boundary Readouts*, QGT Series Paper P11 (2026).

Layer 3 — Cosmology and dimensional closure

- [14] P. Camelia, *Majorana Cosmology*, QGT Series Paper P12, Zenodo (2026).
- [15] P. Camelia, *The Global Phase-Locked Loop: Holonomy-Normalised Coupling Amplitude, Spectral Factor, and the Emergence of Physical Time* (2026).
- [16] P. Camelia, *The Self-Consistent Fine-Structure Constant: Breathing Correction and the Fixed Point of the Observable Boundary*, QGT Series Paper P14, Zenodo (2026).

Companion notes

- [17] P. Camelia, *Structural Ordering from Projection to Time: Finite-Rank Projection, Pre-Physical Distinguishability, and the Emergence of Temporal Order* (2025).
- [18] P. Camelia, *Projection-Induced Decoherence and the Emergence of $1/f$ Noise* (2026).
- [19] P. Camelia, *La fisica come topologia di una proiezione non invertibile* (Italian version of this paper) (2026).